

FUNCTIONS OF ORDER STATISTICS AND TESTS OF FIT

Gavin G. Gregory

The University of Texas at El Paso

Key Words; Cramér-Von Mises test, goodness of fit, limiting distributions, Shapiro-Wilk Statistic, tests of normality, U-statistics, order statistics.

SUMMARY

Let $X_{n1} \leq \dots \leq X_{nn}$ be order statistics of a sample from a continuous distribution. We are concerned with statistics expressible as a differentiable function of several arguments, each of which has the form $\sum_{i=1}^n J_{ni} h(X_{ni})$, where J_{ni} may be generated by $J_{ni} = n^{-1} J(i/(n+1))$. Asymptotic forms for the statistics are found under a basic condition which precludes limiting normality. The asymptotic forms are special U-statistics as treated in Gregory (1977). Applications are limiting distribution for, and asymptotic equivalences among test statistics for goodness of fit to location and/or scale families. Details are given for testing fit to a normal family with zero mean. A Monte Carlo study investigates the approach to the asymptotic distribution.

1. Introduction

Let $X_{n1} \leq \dots \leq X_{nn}$ be order statistics of a sample from a continuous distribution. This paper is concerned with asymptotic theory for a large class of statistics expressible as a differentiable function of several arguments, each of which has the form $\sum_{i=1}^n J_{ni} h(X_{ni})$. The scores J_{ni} may be obtained from a scores generating function by either

$$J_{ni} = \int_{(i-1)/n}^{i/n} J(s) ds \text{ or } J_{ni} = n^{-1} J(i/(n+1)).$$

In Section 2 approximations are obtained for these statistics under conditions which preclude limiting normality. The approximations are special U-statistics to which the results of Gregory (1977) may be applied. The theorems of Section 2 allow unbounded functions J and h but differentiability conditions are freely imposed. Cases are covered where the trace of the kernel of the approximating U-statistics is not finite. Other conditions can be obtained which would apply to functions of frequencies such as chi-square goodness of fit statistics. Our theorems were designed with the applications of Section 3 in mind. The results of Section 2 relate to the work on differentiable statistical functions of Von Mises (1947) and Filippova (1961).

The scope of the applications is indicated in Section 3 and a particular setting is discussed. For testing goodness of fit to a scale parameter family of distributions (with or

without known location parameter) relations are found among test statistics of the Cramér-Von Mises type, those suggested by the Shapiro and Wilk (1965) W statistic and by statistics of De Wet and Venter (1972 and 1973a). We concentrate on testing for normality with mean zero and find a triplet of test statistics, one from each class above, which are asymptotically equivalent under contiguous alternatives. The asymptotic distribution is calculated and a Monte Carlo study is made of the De Wet and Venter type of statistic. This statistic has application in constructing tests of linear model structure, based on the fact that certain transformations on the observations give rise under the model to independent normal variables with zero mean.

This paper is concerned with U-statistics which do not have a limiting normal distribution. However, the techniques used can also produce theorems which conclude limiting normality. The comments at the end of Section 2 suggest that such theorems would have roughly equal force when compared to known results of this kind.

2. General asymptotic approximation results for functions of order statistics which are asymptotically of quadratic type

In this section we consider functions of order statistics $U_{n1} \leq \dots \leq U_{nn}$, of a sample of size n , $U_1, \dots, U_n$, from a uniform distribution on $(0,1)$. The results can be applied to samples from other distributions by utilizing the probability integral transformation. Applications of this type are in Section 3.

We first consider statistics of the form

$$(2.1) \quad T_n = n[G(S_{n1}, \dots, S_{np}) - G(0, \dots, 0)],$$

where

$$S_{nk} = \int_0^1 \{J_k^*[F_n(t)] - J_k^*[t]\} dH_k(t), \quad 1 \leq k \leq p,$$

and $F_n(t)$ is the empirical cdf.

The first result of this section rewrites the S_{nk} . It is convenient to let $J_k^* = J_k$. The following lemma shows that J_k may be called a scores function.

Lemma 2.1

Suppose the following conditions are satisfied for each $J = J_k$ and $H = H_k$, $1 \leq k \leq p$:

- 1) $H(t)$, $0 < t < 1$, is the difference of two monotone non-decreasing functions and $J^*(t)$, $0 \leq t \leq 1$, is absolutely continuous;
- 2) HJ is integrable on $(0,1)$, and

$$\lim_{t \rightarrow 0} H(t) \int_0^t J(s) ds = 0$$

$$\lim_{t \rightarrow 1} H(t) \int_t^1 J(s) ds = 0.$$

Then

$$S_{nk} = \int_0^1 J_k(t) H_k(t) dt - \sum_{i=1}^n \int_{(i-1)/n}^{i/n} J_k(s) ds H_k(U_{ni}).$$

Proof:

We have

$$(2.2) \quad S_{nk} = \int_0^1 \int_t^{F_n(t)} J_k(s) ds dH_k(t).$$

The integrand has discontinuities at $U_{n1}, \dots, U_{nn}$ but with probability one these will not be at discontinuities of H_k . Thus, the above integral may be viewed as a Riemann-Stieltjes integral (see p. 56 of Ash (1972)). Further, (2.2) is

$$(2.3) \quad S_{nk} = \sum_{i=0}^n \int_{U_{ni}}^{U_{n,i+1}} \int_t^{i/n} J_k(s) ds dH_k(t) \quad (U_{n0} = 0, U_{n,n+1} = 1).$$

Integration by parts in (2.3) and collecting terms yields the result. //

As a preliminary to the next theorem we consider a rate condition on functions called Property R, which is an extension of the "reproducing U-shaped function" criterion of Shorack (1972), p. 426.

Property R

A function $\xi(t)$, $0 < t < 1$, has property R if

- 1) ξ is bounded in absolute value on $[\epsilon, 1 - \epsilon]$ for all $0 < \epsilon < 1$, and monotone on $(0, \epsilon)$ and $(1 - \epsilon, 1)$ for some $\epsilon > 0$,
- 2) for each β , $0 < \beta < 1$,

$$\lim_{t \rightarrow 0} |\xi(t)|/|\xi(\beta t)| \neq 0$$

if $|\xi|$ is monotone non-increasing and non-constant on $(0, \epsilon)$.

$$\lim_{t \rightarrow 0} |\xi(t)|/|\xi(t/\beta)| \neq 0$$

if $|\xi|$ is monotone non-decreasing and non-constant on $(0, \epsilon)$.

- 3) $\xi(t) = \xi(1 - t)$ satisfies 2).

Condition 2) becomes important when $\lim_{t \downarrow 0} \xi(t)$ is 0 or $\pm\infty$.

The following theorem gives an asymptotic form for T_n to which the results of

Gregory (1977) may be applied to obtain limiting distributions. Differentiability requirements are placed on the H_k . Conditions allowing discontinuous functions H_k (for instance, indicator functions of intervals) can be obtained by imposing boundedness conditions on the J_k .

Theorem 2.2

Suppose the following conditions are satisfied for each $J = J_k$ and $H = H_k$, $1 \leq k \leq p$:

- 1) J and H satisfy the conditions of Lemma 2.1.
- 2) For every $\epsilon > 0$ suppose that H' is bounded on $[\epsilon, 1 - \epsilon]$.
- 3) J' and J'' have Property R.
- 4) If H' is not bounded near 0(1) then require

$$\lim_{x \rightarrow 0} |H'(x)/x H''(x)| < \infty \quad (\lim_{x \rightarrow 1} |H'(x)/(1-x) H''(x)| < \infty).$$
- 5) The following functions of $t \in (0, 1)$ have zero limits as $t \rightarrow 0$ or $t \rightarrow 1$: $t^2(1-t)^2 J'(t) H'(t)$, $t^3(1-t)^3 J''(t) H'(t)$.
- 6) $[t(1-t)]^{1/2} J(t) H'(t)$, $0 < t < 1$, is integrable.

Suppose $G(S_1, \dots, S_p)$ has continuous second partial derivatives in a neighbourhood of $\underline{0} = (0, \dots, 0)_p$ and write $\lambda_k = (\partial G / \partial S_k)|_{\underline{0}}$ and $\lambda_{k\ell} = (\partial^2 G / \partial S_k \partial S_\ell)|_{\underline{0}}$. If

$$(2.4) \quad \sum_{k=1}^p \lambda_k J_k H'_k \equiv 0$$

then

$$T_n \xrightarrow{P} \hat{T}_n \rightarrow 0,$$

where

$$\hat{T}_n = n^{-1} \sum_{i,j=1}^n Q(U_i, U_j)$$

and

$$(2.5) \quad Q(u, v) = (1/2) \sum_{k=1}^p \lambda_k \int_0^1 \{I[u \leq t] - t\} \{I[v \leq t] - t\} J'_k(t) H'_k(t) dt \\ + (1/2) \sum_{k=1}^p \sum_{\ell=1}^p \lambda_{k\ell} \int_0^1 \{I[u \leq t] - t\} J_k(t) H'_\ell(t) dt \\ + \int_0^1 \{I[v \leq t] - t\} J_\ell(t) H'_k(t) dt, \quad 0 < u, v < 1,$$

where $I[E]$ is the indicator function of the event E .

Proof:

Consider the events

$$E_n(\beta) = \{\beta t < F_n(t) < 1 - \beta(1 - t), U_{n1} \leq t < U_{nn}\}.$$

$$1 - (1 - t)/\beta < F_n(t) < t/\beta, 0 \leq t \leq 1, 0 < \beta < 1.$$

We have

$$\lim_{\beta \rightarrow 0} \liminf_n P(E_n(\beta)) = 1$$

(see Lemma 7.2 of Gregory (1977)). For any r.v. Y_n we will have $Y_n \xrightarrow{P} 0$ if for each β in a neighbourhood of zero, there exists a r.v. X_n^β such that $|Y_n| \leq X_n^\beta$ on $E_n(\beta)$ where $X_n^\beta \xrightarrow{P} 0$. We work henceforth only on $E_n(\beta)$ for an arbitrary positive β near zero.

Let

$$\hat{S}_{nk} = \int_0^1 (F_n(t) - t) J_k(t) H'_k(t) dt, 1 \leq k \leq p.$$

We first show

$$(2.6) \quad n^{1/2} (S_{nk} - \hat{S}_{nk}) = o_p(1)$$

and

$$(2.7) \quad n^{1/2} \hat{S}_{nk} = O_p(1)$$

which implies

$$(2.8) \quad n^{1/2} S_{nk} = O_p(1).$$

We prove (2.6). Since J'_k has Property R there is a finite constant M_β and some $\epsilon > 0$ so that

$$|J'_k(\beta t)| + |J'_k(t/\beta)| \leq M_\beta |J'_k(t)|, 0 < t \leq \epsilon,$$

and

$$|J'_k(1 - \beta(1 - t))| + |J'_k(1 - (1 - t)/\beta)| \leq M_\beta |J'_k(t)|, 1 - \epsilon \leq t < 1.$$

Assume ϵ is small enough so that J'_k is monotone on $(0, \epsilon/\beta]$ and $[1 - \epsilon/\beta, 1)$. We have

$$(2.9) \quad n^{1/2} (S_{nk} - \hat{S}_{nk}) = n^{1/2} \int_0^{U_{n1}} \{J_k^*(0) - J_k^*(t) + t J_k(t)\} H'_k(t) dt$$

$$\begin{aligned}
& + n^{1/2} \left\{ \int_{U_{n1}}^{\epsilon} + \int_{\epsilon}^{1-\epsilon} + \int_{1-\epsilon}^{U_{nn}} \right\} \{J_k^*(F_n(t)) - J_k^*(t) \\
& - (F_n(t) - t) J_k'(t)\} H_k'(t) dt \\
& + n^{1/2} \int_{U_{nn}}^1 \{J_k^*(1) - J_k^*(t) - (1-t) J_k'(t)\} H_k'(t) dt.
\end{aligned}$$

Integrability of J_k implies $t_n J_k(t_n) \rightarrow 0$ for some sequence $\{t_n\}$ with $t_n \downarrow 0$. Utilizing the integration by parts formula for $\int_t^{\epsilon} J_k(s) ds$, where $\epsilon > 0$ is small enough that J_k' is of constant sign on $(0, \epsilon)$, we conclude that $\lim_{t \downarrow 0} t J_k'(t) = 0$. This implies

$$(2.10) \quad J_k^*(0) - J_k^*(t) + t J_k'(t) = \int_0^t s J_k'(s) ds.$$

The first term of (2.9) is shown to be $o_p(1)$ by using (2.10), the relation $U_{n1} \leq 1/\beta n$ holding on $E_n(\beta)$, and the rate conditions of the theorem. Here the rate conditions are used to conclude that $\lim_{t \downarrow 0} \int_0^t s J_k'(s) ds H_k'(t) = 0$. This is obtained by using L'Hospital's

rule with conditions 4) and 5) of the theorem.

The integral over $[U_{n1}, \epsilon]$ in (2.9) is bounded in absolute value by

$$(2.11) \quad n^{1/2} \int_{U_{n1}}^{\epsilon} 2^{-1} (F_n(t) - t)^2 |J_k'(\theta_{nt}) H_k'(t)| dt,$$

where θ_{nt} is between t and $F_n(t)$. Using

$$|J_k'(\theta_{nt})| \leq |J_k'(t)| + |J_k'(F_n(t))| \quad (\text{monotonicity})$$

$$\leq |J_k'(t)| + |J_k'(\beta t)| + |J_k'(t/\beta)|$$

$$\leq (M_{\beta} + 1) |J_k'(t)|,$$

and $U_{n1} \geq \beta/n$, (2.11) is bounded by a non-negative random variable whose expected value approaches zero under the rate condition 5) of the theorem. Thus, in (2.9) the integral over $[U_{n1}, \epsilon]$ is $o_p(1)$. It is clear how to complete the proof of (2.6).

Now the expression in (2.7) is the sum of i.i.d. variables. The result follows from

the Central Limit Theorem since the existence of the variance is a consequence of our assumptions. Thus are (2.6), (2.7) and (2.8) established.

The following statement results from the differentiability assumptions. For each $\epsilon > 0$ there is a $\delta > 0$ so that when $|S_{nk}| < \delta$, $1 \leq k \leq p$,

$$n|G(S_{n1}, \dots, S_{np}) - G(\underline{0}) - \sum_{k=1}^p \lambda_k S_{nk} - (1/2) \sum_{k=1}^p \sum_{\ell=1}^p \lambda_{k\ell} S_{nk} S_{n\ell}| \\ < \epsilon [\max \{n^{1/2}|S_{n1}|, \dots, n^{1/2}|S_{np}|\}]^2.$$

This implies that

$$(2.12) \quad T_n - n \left\{ \sum_{k=1}^p \lambda_k S_{nk} + (1/2) \sum_{k=1}^p \sum_{\ell=1}^p \lambda_{k\ell} S_{nk} S_{n\ell} \right\} \xrightarrow{P} 0.$$

The results (2.6), (2.7) and (2.8) show that

$$(2.13) \quad n(S_{nk} S_{n\ell} - \hat{S}_{nk} \hat{S}_{n\ell}) \xrightarrow{P} 0.$$

The final step in the proof will be showing that

$$(2.14) \quad n \sum_{k=1}^p \lambda_k S_{nk} - n(1/2) \sum_{k=1}^p \lambda_k \int_0^1 (F_n(t) - t)^2 J'_k(t) H'_k(t) dt \xrightarrow{P} 0.$$

Substitution of the approximations implied by (2.13) and (2.14) into (2.12) yields the theorem.

We prove (2.14).

$$(2.15) \quad n \sum_{k=1}^p \lambda_k S_{nk} = n \sum_{k=1}^p \lambda_k \int_0^{U_{n1}} [J_k^*(0) - J_k^*(t)] H'_k(t) dt \\ + n \sum_{k=1}^p \lambda_k \int_{U_{nn}}^1 [J_k^*(1) - J_k^*(t)] H'_k(t) dt \\ + n \sum_{k=1}^p \lambda_k \int_{U_{n1}}^{U_{nn}} [J_k^*(F_n(t)) - J_k^*(t)] H'_k(t) dt.$$

The first term of (2.15) is shown to be $o_p(1)$ by using the expansion (2.10), then (2.4), then the bound $U_{n1} \leq 1/\beta n$ and then the rate conditions of the theorem. The treatment of the second term of (2.15) is similar. Using the Taylor expansion in the last term of (2.15) we conclude that

$$(2.16) \quad n \sum_{k=1}^p \lambda_k S_{nk} - n \sum_{k=1}^p \lambda_k \int_{U_{n1}}^{U_{nn}} (F_n(t) - t)^2 J'_k(t) H'_k(t) dt$$

$$-n \sum_{k=1}^p \lambda_k 6^{-1} \int_{U_{n1}}^{U_{nn}} (F_n(t) - t)^3 J_k''(\tilde{\theta}_{nt}) H_k'(t) dt = o_p(1),$$

where $\tilde{\theta}_{nt}$ is between $F_n(t)$ and t . The last term in the LHS of (2.16) may be bounded by the technique used following (2.11). The expected value of the resulting bound is treated using

$$\begin{aligned} E|F_n(t) - t|^3 &\leq [E|F_n(t) - t|^4]^{3/4} \\ &= \{n^{-2} 3t^2(1-t)^2 + n^{-3} t(1-t)(1-6t+6t^2)\}^{3/4} \\ &\leq \{n^{-2} t^2(1-t)^2(3+1/\beta)\}^{3/4}, \beta/n \leq t \leq 1 - \beta/n. \end{aligned}$$

Thus, it is shown that the last term in the LHS of (2.16) is $o_p(1)$. The result (2.14) then follows since we may easily show

$$n \left\{ \int_0^{U_{n1}} + \int_{U_{nn}}^1 \right\} (F_n(t) - t)^2 J_k'(t) H_k'(t) dt \xrightarrow{P} 0. //$$

Lemma 2.1 shows that the scores in the S_{nk} are the very smooth type $\int_{(i-1)/n}^{i/n} J(s) ds$.

The next theorem concerns the use of the scores $n^{-1} J_k(i/(n+1))$ instead of

$$\int_{(i-1)/n}^{i/n} J(s) ds. \text{ Let}$$

$$\tilde{S}_{nk} = \int_0^1 J_k(t) H_k(t) dt - n^{-1} \sum_{i=1}^n J_k(i/(n+1)) H_k(U_{ni}), 1 \leq k \leq p,$$

and

$$\tilde{T}_n = n[G(\tilde{S}_{n1}, \dots, \tilde{S}_{np}) - G(\underline{0})].$$

Theorem 2.3

Suppose that in addition to the conditions of Theorem 2.2, we have for each $H = H_k$, $1 \leq k \leq p$:

1) H and H' have Property R,

2) $\lim_{x \rightarrow 0} |H(x)/x^{1/2} H'(x)| < \infty$ and $\lim_{x \rightarrow 1} |H(x)/(1-x)^{1/2} H'(x)| < \infty$.

Then

$$\tilde{T}_n - [\hat{T}_n + b_n] \xrightarrow{P} 0,$$

where $\hat{T}_n$ is given in Theorem 2.2 and

$$b_n = n \sum_{k=1}^p \lambda_k \sum_{i=1}^n \left[\int_{(i-1)/n}^{i/n} J_k(s) ds - n^{-1} J_k(i/(n+1)) \right] H_k(i/(n+1)).$$

Proof:

Inspection of the proof of Theorem 2.2 reveals that it is sufficient to show

$$(2.17) \quad n \sum_{k=1}^p \lambda_k (\tilde{S}_{nk} - S_{nk}) - b_n \xrightarrow{P} 0$$

and

$$(2.18) \quad n^{1/2} (\tilde{S}_{nk} - S_{nk}) \xrightarrow{P} 0, 1 \leq k \leq p.$$

The LHS of (2.17) is

$$(2.19) \quad n \sum_{k=1}^p \lambda_k \sum_{i=1}^n \left[\int_{(i-1)/n}^{i/n} J_k(s) ds - n^{-1} J_k(i/(n+1)) \right] [H_k(U_{ni}) - H_k(i/(n+1))].$$

First we show that for each k the terms for $i=1$ and $i=n$ in (2.19) are $o_p(1)$. The term for $i=1$ may be written

$$(2.20) \quad n \left[\int_0^{1/n} J_k(s) ds - n^{-1} J_k(1/(n+1)) \right] (U_{n1} - 1/(n+1)) H'_k(\theta_n),$$

where θ_n is between U_{n1} and $1/(n+1)$. Since H'_k has Property R we have

$|H'_k(\theta_n)| \leq C_\beta |H'_k(1/n)|$ on $E_n(\beta)$ for all large n where C_β is a constant. Now (2.20) may be bounded in absolute value on $E_n(\beta)$ and the expectation of the bound can be shown to approach zero by using $E|U_{n1} - 1/(n+1)| \leq n^{-1}$ and the rate conditions. The term for $i=n$ is also negligible. Thus, (2.17) follows if

$$(2.21) \quad n \sum_{k=1}^p \lambda_k \sum_{i=2}^{n-1} [n^{-2} C_{nik} J'_k(\eta_{nik})] [(U_{ni} - 1/(n+1)) H'_k(\theta_{nik})] = o_p(1),$$

where we have effected the Taylor expansion in (2.19) with

η_{nik} between $(i-1)/n$ and i/n

θ_{nik} between U_{ni} and $i/(n+1)$,

and $|C_{nik}| \leq 1$.

We show (2.21) by bounding the LHS in absolute value on $E_n(\beta)$ and showing the expectation of the bound approaches zero. For $2 \leq i \leq n-1$ we have $i/2(n+1) \leq \eta_{nik} \leq 1 - (1 - i/(n+1))/2$. On $E_n(\beta)$, $\beta i/(n+1) \leq U_{ni} \leq 1 - \beta(1 - i/(n+1))$. Also

$$(2.22) \quad E|U_{ni} - i/(n+1)| \leq [(i/(n+1))(1 - i/(n+1))/(n+2)]^{1/2}$$

Since J'_k and H'_k are bounded on any region excluding neighbourhoods of zero and one, components of (2.21) involving summation over i , $[n\epsilon] < i < [n(1-\epsilon)]$ for arbitrary $\epsilon > 0$, need not be considered. Choose ϵ so small that for the remaining indices i , and on $E_n(\beta)$, the arguments of J'_k and H'_k are restricted to a region near zero and a region near one on which the functions are monotone. Since J'_k and H'_k have Property R, (2.21) follows from

$$(2.23) \quad n^{-3/2} \sum_{\substack{[n(1-\epsilon)] \leq i \leq n-1 \\ 2 \leq i \leq [n\epsilon]}} |J'_k(i/(n+1))| [i/(n+1)(1 - i/(n+1))]^{1/2} \\ H'_k(i/(n+1)) = o(1), \quad 1 \leq k \leq p.$$

The monotonicity property of the functions in (2.23) may be invoked to get that (2.23) follows from

$$(2.24) \quad n^{-1/2} \int_{1/n}^{1-1/n} |J'_k(t)(t(1-t))^{1/2} H'_k(t)| dt = o(1).$$

Now (2.24) is easy and therefore (2.17) is obtained.

The LHS of (2.18) is

$$n^{1/2} (\tilde{S}_{nk} - S_{nk}) = n^{1/2} \sum_{i=1}^n \left[\int_{(i-1)/n}^{i/n} J_k(s) ds - n^{-1} J_k(i/(n+1)) \right] H_k(U_{ni}).$$

The proof is completed using the techniques that were applied to (2.19). //

By looking at the form of b_n we expect that

$$b_n \rightarrow \sum_{k=1}^p \lambda_k \int_0^1 J'_k(s) H_k(s) ds$$

if $J'_k H_k$ is integrable. Integration by parts and using (2.4) shows $b_n \rightarrow 0$ if $J_k(s) H_k(s)$,

$0 < s < 1$, has zero limit as $s \rightarrow 0$ or 1 . These conditions are more restrictive than we have

assumed and for applications in Section 3 they do not hold. When it cannot be concluded that $b_n \rightarrow 0$ one would like a more revealing form for b_n than the one given in Theorem 2.3 but we were able to accomplish this only for some examples (see Section 3).

Final results discuss the behaviour of the component.

$$(2.25) \quad n^{-1} \sum_{i=1}^n Q(U_i, U_i)$$

of $\hat{T}_n$ (Theorem 2.2). The rest of $\hat{T}_n$ has a limiting distribution with zero mean. Now (2.25) approaches $\int_0^1 Q(u, u) du$ if the integral exists. Our conditions do not ensure the existence of the integral. The difficulty is with the terms of the first summation in Q of (2.5). Since terms like these have independent interest we state a general lemma and then apply it to (2.25).

Lemma 2.4

Suppose w is a differentiable function on $(0, 1)$ and let $h(t) = t(1-t)w(t)$, $0 < t < 1$.

If

- 1) h' has Property R and $h'(t)[t(1-t)]^{1/2}$ is monotone for t near 0 and 1, and
- 2) w' has Property R and the following functions of $t \in (0, 1)$ have zero limits as $t \rightarrow 0$ or $t \rightarrow 1$:
 $t^2(1-t)^2 w(t)$, $t^3(1-t)^3 w'(t)$,

then

$$\begin{aligned} & \int_0^1 n^{-1} \sum_{i=1}^n \{I[U_i \leq t] - t\}^2 w(t) dt \\ & - (n+1)^{-1} \sum_{i=1}^n (i/(n+1))(1-i/(n+1)) w(i/(n+1)) \xrightarrow{P} 0. \end{aligned}$$

Proof:

$$\text{Use } n^{-1} \sum_{i=1}^n \{I[U_i \leq t] - t\}^2 = F_n(t)(1 - F_n(t)) + (F_n(t) - t)^2$$

to write

$$\begin{aligned} (2.26) \quad & \int_0^1 n^{-1} \sum_{i=1}^n \{I[U_i \leq t] - t\}^2 w(t) dt \\ & = \int_{U_{n1}}^{U_{nn}} F_n(t)(1 - F_n(t)) w(F_n(t)) dt \\ & + \int_{U_{n1}}^{U_{nn}} F_n(t)(1 - F_n(t)) (w(t) - w(F_n(t))) dt \end{aligned}$$

$$+ \int_0^1 (F_n(t) - t)^2 w(t) dt.$$

Only the first term of (2.26) is treated; the others can be shown to be $o_p(1)$ by the techniques already used. The first term of (2.26) may be written

$$\begin{aligned} (2.27) \quad & (n+1)^{-1} \sum_{i=1}^n (i/(n+1))(1-i/(n+1)) w(i/(n+1)) \\ & + (n+1)^{-1} \sum_{i=1}^{n-1} (1/n)(1-i/n) w(i/n) \\ & - (n+1)^{-1} \sum_{i=1}^n (i/(n+1))(1-i/(n+1)) w(i/(n+1)) \\ & + \sum_{i=1}^{n-1} (i/n)(1-i/n) w(i/n) [(U_{n,i+1} - (i+1)/(n+1)) - (U_{n,i} - i/(n+1))]. \end{aligned}$$

The last term of (2.27) may be written

$$\begin{aligned} (2.28) \quad & \sum_{i=2}^{n-1} [h((i-1)/n) - h(i/n)] (U_{ni} - i/(n+1)) \\ & - h(1/n) (U_{n1} - 1/(n+1)) + h(1-1/n) (U_{nn} - n/(n+1)). \end{aligned}$$

Using (2.22) and by now familiar technique we can finish the proof. //

Applying Lemma 2.4 to (2.25) we get

Lemma 2.5

Suppose for k , $1 \leq k \leq p$, $J'_k H'_k$ satisfies the conditions for w of Lemma 2.4 and that the integrals below exist. Then

$$n^{-1} \sum_{i=1}^n Q(U_i, U_i) - a_n \xrightarrow{P} 0,$$

where Q is given by (2.5) and

$$\begin{aligned} a_n = & 2^{-1} \sum_{k=1}^p \lambda_k (n+1)^{-1} \sum_{i=1}^n (i/(n+1))(1-i/(n+1)) J'_k(i/(n+1)) H'_k(i/(n+1)) \\ & + 2^{-1} \sum_{k=1}^p \sum_{\ell=1}^p \lambda_{k\ell} \int_0^1 \int_0^1 \{s \wedge t - st\} J_k(s) H'_k(s) J_\ell(t) H'_\ell(t) ds dt. \end{aligned}$$

The integrals in a_n above exist under the conditions of Theorem 2.2.

As an aid in determining the integrals we present

Lemma 2.6

If h_1 and h_2 are each the difference of two continuous monotone functions on $(0,1)$ then

$$\int_0^1 h_1 h_2 - \int_0^1 h_1 \cdot \int_0^1 h_2 = \int_0^1 \int_0^1 \{u \wedge v - uv\} h'_1(u) h'_2(v) du dv$$

if the integrals on the left exist.

Proof:

We may assume h_1 and h_2 are monotone increasing. Apply Proposition 5 of Stigler (1974) for $X = h_1(U)$ and $Y = h_2(U)$ with U having a uniform $(0,1)$ distribution. //

Finally, it is appropriate to comment on the type of results that could be obtained if our methods were applied to statistics which were of asymptotically normal type. These types would be found if the degeneracy condition (2.4) were abandoned. The normalizing factor in T_n would be changed from n to $n^{1/2}$. Theorem 2.2 would conclude the equivalence of T_n to the sum of i.i.d. random variables and the major change in the conditions would be that the functions in condition 5) would be replaced by $t^{5/2} (1-t)^{5/2} J'(t) H'(t)$. Theorem 2.3 would allow different scores to be used and condition 2) could be relaxed. Shorack (1972) proves limiting normality for single functions S_{nk} (our notation). The rate conditions of his Theorem 2 seem to have about the same force as the ones we could develop but ours would involve more higher order derivatives. His condition 6a is stronger than condition 6) of Theorem 2.2. This condition 6) (which ensures the existence of the variance of $\hat{S}_{nk}$ as well as integrals in Lemma 2.5) itself can be weakened, both in the results we present and their modification when (2.4) does not hold.

3. Applications to tests of fit

The W statistic for normality was introduced by Shapiro and Wilk (1965). Pertinent to this statistic are Shapiro and Francia (1972), Weisberg and Bingham (1975) and David (1970), pp. 218-222. Van der Watt (1969) indicated the non-normal limiting distribution of the W statistic. De Wet and Venter (1972) consider an appealing test statistic for normality and relate it to a W statistic which uses asymptotic forms for the scores. De Wet and Venter (1973a) consider statistics for testing fit to a general scale parameter family of distributions. De Wet and Venter (1973b) contains 1) distribution theory used in their papers cited above and 2) a discussion of a variant of the classical Cramér-Von Mises statistics for uniformity.

The statistics in the De Wet and Venter papers may be treated by our theorems. How this is done will be clear from consideration of the following problem.

Suppose we have order statistics $X_{n1} \leq \dots \leq X_{nn}$ of a sample of size n from an unknown cdf F . Let F_0 be a specified cdf and suppose we desire to test either $H_1 : F(x) =$

$F_0(x/\theta)$, for some $\theta > 0$ or $H_2: F(x) = F_0((x - \mu)/\theta)$, for some $\theta > 0$ and real μ . We consider tests only of H_1 ; statistics for testing H_2 may be generated from those for testing H_1 by replacing X_{ni} by $X_{ni} - \bar{X}$. For testing H_1 we consider three classes of test statistics:

1) Cramér-Von Mises statistics as discussed in Gregory (1977),

$$(3.1) \quad 2) \quad L_n = \sum_{i=1}^n [X_{ni}/\hat{\theta} - F_0^{-1}(i/(n+1))]^2 w(i/(n+1)), \text{ and}$$

$$(3.2) \quad 3) \quad W_n = n \left\{ 1 - \left[\sum_{i=1}^n n^{-1} F_0^{-1}(i/(n+1)) w(i/(n+1)) X_{ni} \right]^2 \right. \\ \left. \cdot \left[\sum_{i=1}^n n^{-1} w(i/(n+1)) X_{ni}^2 \right]^{-1} \left[\int_0^1 (F_0^{-1})^2 w \right]^{-1} \right\}$$

where $w(t)$, $0 < t < 1$, is a weight function. In each case the hypothesis is rejected for large values of the statistic. Cramér-Von Mises statistics are intuitively appealing but are not always easy to calculate. The form L_n is of the type considered in De Wet and Venter (1973a) and has intuitive appeal as well as calculational simplicity. We introduce the class W_n which may be called generalized W statistics. When X_{ni} is replaced by $X_{ni} - \bar{X}$ and $w \equiv 1$ we have the large sample form of the Shapiro-Wilks W statistic. It can be demonstrated why any statistic W_n with positive weight function will provide a consistent test of H_1 against any alternative, but the motivation behind these statistics is not as immediate as that for L_n . Now the results of Section 2 may be applied to find asymptotic forms for L_n and W_n . Below we treat only the form L_n . At the end of the section we present a result relating the three classes of statistics.

Consider the statistics L_n (3.1) when

$$\hat{\theta} = \left[\sum_{i=1}^n n^{-1} X_{ni}^2 \right]^{1/2}.$$

For convenience assume that the weight function w is normalized so that $\int_0^1 (F_0^{-1})^2 w = 1$.

We desire to discuss L_n when the hypothesis H_1 is true. Since the distribution of L_n under H_1 does not depend on θ assume for convenience that $\theta = 1$ ($\int_0^1 (F_0^{-1})^2 = 1$). Then

we can represent the order statistics X_{ni} as $F_0^{-1}(U_{ni})$ where the U_{ni} are as in Section 2. Squaring out the term in (3.1) and using the notation of Section 2 it is easy to show that L_n may be expressed as

$$(3.3) \quad L_n = n \{G(\tilde{S}_{n1}, \tilde{S}_{n2}, \tilde{S}_{n3}) - G(0)\} + c_n,$$

where

$$(3.4) \quad c_n = n \left\{ \sum_{i=1}^n n^{-1} [F_0^{-1}(i/n + 1)]^2 w(i/(n + 1)) - 1 \right\}$$

$$G(S_1, S_2, S_3) = 1 + [1 - S_3 - 2(1 - S_1)^{1/2} (1 - S_2)] / (1 - S_1)$$

$$J_1 \equiv 1, \quad H_1 = (F_0^{-1})^2$$

$$J_2 = F_0^{-1} w, \quad H_2 = F_0^{-1}$$

$$J_3 = w, \quad H_3 = (F_0^{-1})^2.$$

Calculation of the partial derivatives of G shows that

$$\lambda_1 = 0, \lambda_2 = 2, \lambda_3 = -1$$

and

$$\lambda_{11} = 1/2, \lambda_{12} = 1, \lambda_{13} = -1, \lambda_{22} = \lambda_{23} = \lambda_{33} = 0.$$

The basic condition (2.4) is satisfied and for sufficiently well-behaved functions F_0^{-1} and w the theorems of Section 2 can be applied.

Before applying the theorems, specialize to the case $F_0 = \Phi$ and $w \equiv 1$, where Φ is the standard normal cdf. Let ϕ be the standard normal density. With this choice for w we have $S_{n1} = S_{n3}$ and we could have expressed L_n as a function of two variables. To show that the rate conditions of Theorems 2.2 and 2.3 and Lemma 2.5 are satisfied, make use of

$$(3.5) \quad \lim_{t \rightarrow 0} \text{or } 1 |\Phi^{-1}(t)| (-2 \log t(1-t))^{-1/2} = 1$$

$$\lim_{t \rightarrow 0} \text{or } 1 t(1-t) (-2 \log t(1-t))^{1/2} / \phi \Phi^{-1}(t) = 1.$$

Details are omitted. The kernel Q (2.5) is found to be

$$(3.6) \quad Q(u, v) = \int_0^1 \{I[u \leq t] - t\} \{I[v \leq t] - t\} (1/\phi \Phi^{-1}(t))^2 dt$$

$$- \int_0^1 \{I[u \leq t] - t\} (\Phi^{-1}(t)/\phi \Phi^{-1}(t)) dt$$

$$\cdot \int_0^1 \{I[v \leq t] - t\} (\Phi^{-1}(t)/\phi \Phi^{-1}(t)) dt, \quad 0 < u, v < 1.$$

The theorems of Section 2 show that L_n (3.3) satisfies

$$(3.7) \quad L_n - c_n - b_n - a_n - n^{-1} \sum_{i \neq j} Q(U_i, U_j) \xrightarrow{P} 0,$$

where b_n and a_n are in Theorem 2.2 and Lemma 2.5 respectively, and c_n is given by (3.4) with $F_0 = \Phi$ and $w \equiv 1$.

In Section 5 of Gregory (1977) is studied a Cramér-Von Mises statistic for determining fit to a normal distribution with unknown mean and variance. It is easy to modify the discussion for the case when only the variance is unknown and the mean is specified to be zero. Then the kernel Q found there would, when transformed to the unit interval, be (3.6). We conclude firstly that the test based on L_n is asymptotically equivalent under contiguous alternatives to a Cramér-Von Mises test and secondly that

$$L_n - c_n - b_n - a_n \xrightarrow{D} \sum_{\substack{k=1 \\ k \neq 2}}^{\infty} (Z_k^2 - 1)/k,$$

where $Z_1, Z_2, \dots$ are i.i.d. standard normal variables. Substituting into the formula for b_n in theorem 2.3 it is found that

$$(3.8) \quad c_n + b_n = -n \sum_{i=1}^n \int_{(1-i)/n}^{i/n} [\Phi^{-1}(s) - \Phi^{-1}(i/(n+1))]^2 ds.$$

After showing that the terms for $i=1$ and $i=n$ in the RHS of (3.8) are $o(1)$, the remainder is shown to be $o(1)$ by using techniques in the proof of Theorem 2.3. Now a_n of Lemma 2.5 may be calculated with the help of Lemma 2.6. It is found that

$$a_n = (n+1)^{-1} \sum_{i=1}^n (i/(n+1)) (1 - i/(n+1)) [1/\phi \Phi^{-1}(i/(n+1))]^2 - 1/2.$$

We have established

Theorem 3.1

Under $H_1 : F(x) = \Phi(x/\theta)$, for some $\theta > 0$,

$$L_n - a_n \xrightarrow{D} \sum_{\substack{k=1 \\ k \neq 2}}^{\infty} (Z_k^2 - 1)/k,$$

where $F_0 = \Phi$ and $w \equiv 1$ in (3.3).

This result is the companion result to Theorem 2 of De Wet and Venter (1972) for the case where the location parameter is known. Their a_n^0 appears in our constant as well as the limit $(1/2)$ of their a_n'' .

For testing normality with a specified mean the asymptotic equivalence of a member

of the class L_n (3.1) to a Cramér-Von Mises statistic has been demonstrated. Indeed, it can be found that the kernels Q generated by arbitrary weight functions in (3.1) is the same class of kernels that are generated by the Cramér-Von Mises statistics for H_1 . The kernels for the statistic W_n (3.2) can be found but this class of kernels is *not* the same as the Cramér-Von Mises class. However, for constant weight function, L_n and W_n are asymptotically equivalent. We state without proof

Theorem 3.2

Under $H_1 : F(x) = \Phi(x/\theta)$, for some $\theta > 0$, the statistic L_n (3.1) with weight function w is asymptotically equivalent to the Cramér-Von Mises statistic with weight function $w(u)/\phi^2 \Phi^{-1}(u)$, $0 < u < 1$ (see Gregory (1977)). L_n (3.1) and W_n (3.2) are asymptotically equivalent under H_1 when $w \equiv 1$.

A similar result holds when the mean is not specified by the hypothesis.

A numerical study of the convergence in Theorem 3.1 was carried out. Points on the limiting distribution were obtained by a numerical integration following the method of De Wet and Venter (1972 and 1973a). Points on the distribution of $L_n - a_n$ were estimated by Monte Carlo simulation using the Box-Muller transformation and a dual generator for the uniform variables following Chen (1971). The cdf values are presented in Tables 1 and 2 for various choices of arguments x and sample sizes n . Table 2 shows that a value near $x = 2$ appears to be a stable point, with convergence occurring from above in a long tail to the right of $x = 2$ and occurring from below for positive values to the left of $x = 2$. For percentiles above the 90th the asymptotic values provide a good fit for $n = 30$ samples, a reasonably good fit for $n = 20$ samples, and perhaps even a useful fit for $n = 10$ samples.

TABLE 1

$$y = P \left\{ \sum_{\substack{k=1 \\ k \neq 2}}^{\infty} (Z_k^2 - 1)/k \leq x \right\}$$

x	y	x	y	x	y
0,0	0,604	2,2	0,909	4,4	0,976
0,1	0,629	2,3	0,914	4,5	0,977
0,2	0,652	2,4	0,919	4,6	0,979
0,3	0,675	2,5	0,924	4,7	0,980
0,4	0,697	2,6	0,929	4,8	0,981
0,5	0,716	2,7	0,933	4,9	0,982
0,6	0,735	2,8	0,937	5,0	0,983
0,7	0,752	2,9	0,941	5,1	0,984
0,8	0,769	3,0	0,945	5,2	0,985
0,9	0,784	3,1	0,948	5,3	0,986
1,0	0,798	3,2	0,951	5,4	0,987-
1,1	0,811	3,3	0,954	5,5	0,987+
1,2	0,823	3,4	0,957	5,6	0,988
1,3	0,835	3,5	0,959	5,7	0,989-
1,4	0,846	3,6	0,962	5,8	0,989+
1,5	0,856	3,7	0,964	5,9	0,990-
1,6	0,865	3,8	0,966	6,0	0,990+
1,7	0,874	3,9	0,968	6,5	0,991
1,8	0,882	4,0	0,970	7,0	0,995
1,9	0,889	4,1	0,971	8,0	0,997
2,0	0,896	4,2	0,973	9,0	0,998
2,1	0,903	4,3	0,975		

TABLE 2

Cumulative Distribution Function Values for $L_n - a_n$
Based on 8 000 (5 000) Iterations for $n \leq 20$ ($20 < n \leq 50$)

$\frac{x}{n}$	.2	.4	.6	.8	1.0	1.2	1.4	1.6	1.8	2.0	2.2	2.4	2.6	2.8	3.0	3.2	3.4	3.6	3.8	4.0	4.2	4.4	4.6	4.8	5.0
2	.277	.418	.528	.622	.712	.797	.873	.946	1.0																
4	.429	.544	.637	.703	.758	.801	.841	.875	.901	.925	.944	.958	.969	.978	.987	.990	.995	.997	.998	.999	1.0	1.0			
6	.484	.580	.651	.714	.763	.801	.837	.864	.887	.906	.921	.939	.952	.961	.968	.975	.981	.985	.987	.989	.991	.993	.994	.996	.997
8	.511	.595	.665	.722	.767	.800	.830	.859	.880	.901	.917	.931	.940	.951	.960	.969	.973	.978	.982	.986	.989	.991	.994	.995	.995 ⁺
10	.533	.614	.678	.726	.767	.805	.835	.863	.881	.898	.914	.928	.938	.949	.957	.963	.969	.975	.980	.982	.985	.988	.990	.992	.993
12	.541	.615	.675	.730	.771	.807	.837	.861	.881	.898	.911	.922	.933	.943	.953	.960	.966	.972	.976	.979	.982	.985	.987	.989	.990
14	.550	.627	.680	.733	.775	.807	.835	.860	.878	.895	.909	.921	.931	.941	.950	.958	.965	.971	.974	.978	.981	.983	.986	.988	.990
16	.557	.632	.691	.736	.778	.809	.837	.861	.879	.897	.912	.924	.935	.945	.952	.959	.963	.970	.974	.978	.981	.984	.985	.987	.989
18	.568	.635	.693	.734	.773	.810	.837	.863	.882	.897	.911	.923	.934	.943	.950	.958	.964	.970	.974	.978	.982	.984	.987	.989	.990
20	.566	.632	.690	.735	.775	.805	.838	.860	.880	.895	.912	.924	.936	.943	.951	.957	.963	.967	.972	.977	.981	.983	.985	.988	.989
22	.597	.642	.693	.733	.772	.805	.835	.861	.879	.893	.911	.921	.930	.938	.946	.953	.960	.967	.969	.973	.976	.979	.982	.985	.987
24	.576	.640	.691	.739	.776	.809	.836	.864	.881	.897	.909	.921	.930	.939	.948	.955	.960	.964	.969	.973	.975	.979	.981	.984	.985
26	.581	.647	.699	.744	.778	.807	.835	.859	.878	.895	.908	.920	.931	.937	.944	.951	.958	.963	.967	.971	.974	.977	.982	.984	.988
28	.581	.640	.692	.735	.774	.804	.833	.858	.877	.893	.906	.919	.927	.937	.944	.951	.956	.961	.965	.971	.976	.979	.981	.984	.986
30	.586	.646	.697	.735	.773	.805	.833	.858	.877	.892	.907	.919	.931	.939	.946	.951	.957	.962	.965	.971	.975	.978	.981	.984	.986
32	.590	.650	.693	.735	.776	.804	.832	.855	.875	.893	.910	.921	.929	.938	.945	.954	.960	.966	.968	.972	.975	.978	.982	.985	.987
34	.588	.639	.691	.739	.770	.803	.834	.858	.880	.892	.905	.920	.930	.939	.946	.953	.959	.964	.969	.973	.976	.979	.982	.985	.988
36	.590	.650	.695	.743	.775	.806	.833	.858	.876	.891	.908	.920	.926	.936	.942	.949	.956	.962	.966	.972	.976	.978	.981	.983	.985
38	.594	.655	.705	.741	.773	.804	.829	.853	.879	.895	.906	.918	.928	.938	.945	.951	.958	.963	.969	.973	.978	.979	.982	.983	.984
40	.600	.655	.703	.745	.780	.811	.835	.857	.875	.892	.909	.918	.929	.938	.947	.953	.959	.962	.968	.972	.976	.978	.981	.983	.985
42	.596	.654	.702	.744	.780	.811	.837	.858	.877	.891	.904	.916	.927	.935	.944	.950	.957	.963	.968	.973	.977	.980	.983	.984	.986
44	.602	.663	.706	.743	.778	.807	.832	.857	.875	.892	.904	.916	.926	.933	.943	.951	.957	.962	.968	.973	.976	.980	.982	.984	.985
46	.598	.658	.706	.745	.778	.810	.840	.859	.880	.895	.909	.923	.931	.938	.946	.954	.960	.966	.969	.973	.976	.978	.980	.982	.985
48	.602	.657	.702	.742	.777	.807	.835	.859	.878	.895	.906	.918	.928	.941	.947	.953	.959	.963	.968	.971	.974	.977	.979	.983	.985
50	.604	.662	.708	.746	.778	.808	.836	.859	.878	.893	.907	.919	.928	.940	.947	.953	.958	.963	.967	.973	.977	.979	.982	.984	.986
∞	.653	.697	.735	.769	.798	.823	.846	.865	.882	.896	.909	.919	.927	.937	.945	.951	.957	.961	.966	.970	.973	.976	.979	.981	.983

Acknowledgment

Helpful comments by J.H. Venter aided in preparing the manuscript.

REFERENCES

- ASH, R.B. (1972). *Real Analysis and Probability*. Academic Press, New York.
- CHEN, E.H. (1971). Random normal number generator for 32-bit-word computers. *J. Amer. Statist. Assoc.* **66** 400-403.
- DAVID, H.A. (1970). *Order Statistics*. Wiley, New York.
- DE WET, T. and VENTER, J.H. (1972). Asymptotic distributions of certain test criteria of normality. *South African Statist. J.* **6** 135-149.
- DE WET, T. and VENTER, J.H. (1973a). A goodness of fit test for a scale parameter family of distributions. *South African Statist. J.* **7** 35-46.
- DE WET, T. and VENTER, J.H. (1973b). Asymptotic distributions for quadratic forms with applications to tests of fit. *Ann. Statist.* **1** 380-387.
- FILIPPOVA, A.A. (1961). Mises theorem on the asymptotic behavior of functionals of empirical distribution functions and its statistical applications. *Theor. Probability Appl.* **7** 24-56.
- GREGORY, G.G. (1977). Large sample theory for U-statistics and tests of fit. *Ann. Statist.* **5** 110-123.
- SHAPIRO, S.S. and WILKS, M.B. (1965). An analysis of variance test for normality (complete samples). *Biometrika* **52** 591-611.
- SHAPIRO, S.S. and FRANCIA, R.S. (1972). An approximate analysis of variance test for normality. *J. Amer. Statist. Assoc.* **67** 215-216.
- SHORACK, G.R. (1972). Functions of order statistics. *Ann. Math. Statist.* **43** 412-427.
- STIGLER, S.M. (1974). Linear functions of order statistics with smooth weight functions. *Ann. Statist.* **2** 676-693.
- VAN DER WATT, P. (1969). A comparison of certain tests of normality. *South African Statist. J.* **3** 69-82.
- VON MISES, R. (1947). On the asymptotic distribution of differentiable statistical functions. *Ann. Math. Statist.* **18** 309-348.
- WEISBURG, S. and BINGHAM, C. (1975). An approximate analysis of variance test for non-normality suitable for machine calculation. *Technometrics* **17** 133-136.

Manuscript received 1977-03-09, revised 1977-05-24.